When Capability Creates Opportunity

Vincent Boucher — President of MONTREAL.AI

27 October 2026 · Portable production study · Edition 01

Decision

Retain the working engineering artifact. Stop the tested formulation as the transformational flagship. Hold commercial admission pending an actual buyer-defined need. The five-candidate tournament did not establish a new productive opportunity that survives its strongest conventional alternative.

The investigated possibility was portable construction: turn product requirements into an executable capability that another competent producer can use and revise, potentially making previously uneconomic work feasible. The missing evidence is a buyer-relevant production boundary crossed by that capability. Successful nominal geometry does not supply it.

What is real

The package contains an executable nominal-CAD constructor, editable feature inputs, STEP/STL/SVG artifacts, feature tables and a reproducible checker. Three public CadQuery educational examples were reconstructed. Three new researcher-specified geometric requests were then tested after the constructor froze.

Evidence	Observed result
Original frozen run	0/3 accepted; failure preserved
Corrected historical reproduction	3/3 accepted
Fresh post-freeze nominal requests	3/3 accepted
Predeclared geometric rejection cases	6/6 refused
Additional missing/duplicate feature-ID cases	0/2 refused; limitation retained
Supplementary STEP geometry comparisons	6/6 equivalent

The original failure involved bounding-box measurement after tessellation. A separately recorded amendment corrected that measurement without weakening the criterion. A later review also repaired a supplementary verifier that could trust empty detailed evidence. Both failures and their corrections remain inspectable. [E2–E3]

Why this does not admit the flagship

The free native CadQuery source already creates the tested geometry. It also retains dimensional relationships that the new explicit-coordinate recipe loses. Direct modeling can edit imported geometry without reconstructing its history; CAD-Recode and established feature-recognition tools further constrain novelty. [1–3]

The examples are educational, not commissioned industrial work. The recipe does not infer material, tolerances, design intent, machine instructions or authority. Both geometric paths use the same OpenCascade kernel. No customer accepted an outcome, and no physical part was produced.

The commercial proposal and its conditional economics

The modeled future offering is a **source-bound nominal CAD conformance/test artifact with explicitly supplied variants**, proposed at **USD1,200 earned on final acceptance**. Terms include a 50% refundable restricted deposit, one correction, a full refund on failure, the remaining balance thirty days after acceptance and costed support. This is a reviewable proposal, not a live offer, quote or sale.

All production-cost and demand assumptions are exposed. Loaded labor is assumed at USD100/hour; the BLS wage anchor is USD50.05/hour for mechanical engineers in May 2025. [15] The model counts acquisition, preparation, delivery, validation, correction, failed work, payment costs, support, default and coordination.

Shared preparation, USD1,200 fee	First only	Selective continuation
Public-source zero-value control	−75.00	−75.00
Adverse custom-work assumptions	−110.17	−110.17
Central custom-work assumptions	−275.11	−252.59
Favorable custom-work assumptions	−235.64	+146.98

The public-source zero-value control assumes no incremental buyer value; no buyer refusal was observed. Stopping earns zero incremental profit. The favorable selective policy still loses in **37.75%** of modeled histories. A separate illustrative USD2,400 historical build charge reduces its result to **USD−2,253.02**. That charge is not a measurement of this study's cost.

The conventional shared provider and proposed shared provider tie in all twelve matched scenario/policy cells because they receive the same asset, method and information. The scenario isolates ordinary sharing; it does not measure Operational Alpha. The 7,122 weighted histories are mathematical cases, not buyer observations. [E4]

Cash and scale

The first booked attempt requires **USD1,542 unrestricted initial capital** under the frozen worst-case reservation. The refundable deposit cannot finance that requirement. A fully collected clean first delivery with one acquisition loses **USD139** at the base fee; one correction increases the loss to **USD294**. The modeled clean first-delivery break-even is **USD1,341.84**. [E4–E5]

The scale ambition is explicitly **USD1 billion cumulative net customer benefit over ten years**, not revenue or valuation. Using the Census population anchor of 17,163 machine-shop establishments, an assumed USD250 net benefit per accepted outcome and the frozen adoption ramp, the target requires four million accepted outcomes—approximately 203.55 per active shop per year. Twenty per year yields USD98.26 million. Eligibility, adoption and actual value remain unvalidated. [16,E4]

The next decision

Prepare one buyer-defined case with a real unavailable outcome, authorized inputs, acceptance criteria, outside options and a binding cost or deadline. Give the strongest conventional challenger that same problem. Proceed only if a capability gap survives and the first artifact preserves the essential transformation.

No outreach, new spending commitments, financial transactions, machine operation or live authentication was performed. Actual customers, commitments and collections are zero; actual full study cost and profit are unknown. Operational Alpha is unmeasured. Additional retention beyond an identical ordinary library is rejected in the scenario; Gym training was not run.

The full paper includes the five-candidate tournament, complete prior-art comparison, staged decision model, evidence map, versioned failures and reproducibility instructions. References [1–22] and evidence families [E1–E6] correspond to the complete paper. This AI-assisted research draft is not peer-reviewed acceptance or production deployment.

FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.